EXAMPLE 7.9

Light of wavelength 500 nm passes through a slit of 0.2 mm wide. The diffraction pattern is formed on a screen 60 cm away. Determine the,

(a) angular spread of central maximum

(b) the distance between the central maximum and the second minimum.

Solution

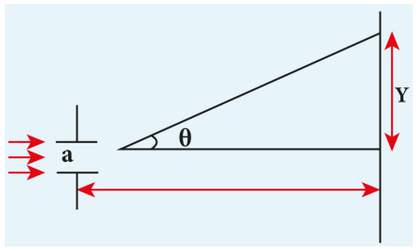
λ = 500 nm = 500×10^{-9} m

a = 0.2 mm = 0.2×10^{-3} m

D = 60 cm = 60×10^{-2} m

(a) Equation for diffraction minimum is, $a \sin\theta = n\lambda$

The central maximum is spread up to the first minimum. Hence, $n = 1$



Rewriting, $\sin\theta = \dfrac{\lambda}{a}$ (or) $\theta = \sin^{-1}\left(\dfrac{\lambda}{a}\right)$

Substituting,

$$\theta = \sin^{-1}\left(\frac{500\times10^{-9}}{0.2\times10^{-3}}\right) = \sin^{-1}\left(2.5\times10^{-3}\right)$$

$\theta = 0.0025$ rad

(b) To find the value of y_1 from the central maximum, which is spread up to first minimum with $(n = 1)$,

$a \sin\theta = \lambda$

As θ is very small, $\sin\theta \approx \tan\theta = \dfrac{y_1}{D}$

$$a\frac{y_1}{D} = \lambda \qquad \text{rewriting, } y_1 = \frac{\lambda D}{a}$$

Substituting,

$$y_1 = \frac{500\times10^{-9}\times60\times10^{-2}}{0.2\times10^{-3}} = 1.5\times10^{-3} = 1.5 \text{ mm}$$

To find the value of y_2 for second minimum with $(n = 2)$,

$a \sin\theta = 2\lambda$

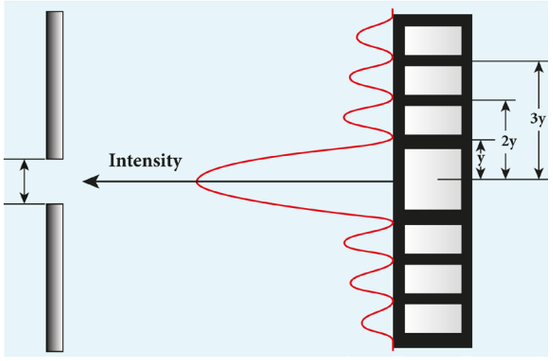
$$a\frac{y_2}{D} = 2\lambda \qquad \text{rewriting, } y_2 = \frac{2\lambda D}{a}$$

Substituting,

$$y_2 = \frac{2\times500\times10^{-9}\times60\times10^{-2}}{0.2\times10^{-3}} = 3\times10^{-3} = 3 \text{ mm}$$

The distance between the central maximum and second minimum is, $y_2 - y_1$

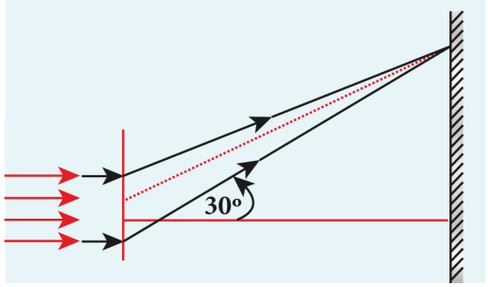
$y_2 - y_1$ = 3 mm – 1.5 mm = 1.5 mm



Note: The above calculation shows that the diffraction pattern produced by a single slit, has equal widths of maxima. Only the width of central maximum is double as it is spread on both the sides. But, the intensity falls rapidly for higher order diffraction fringes.

EXAMPLE 7.10

A monochromatic light of wavelength 5000 Å passes through a single slit producing diffraction pattern for the central maximum as shown in the figure. Determine the width of the slit.



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